Section A

Q1. (1 Mark) What is a backdoor attack in the context of safety of AI systems?

Q2. (1 Mark) Consider the 0–1 loss (classification error loss) defined as:

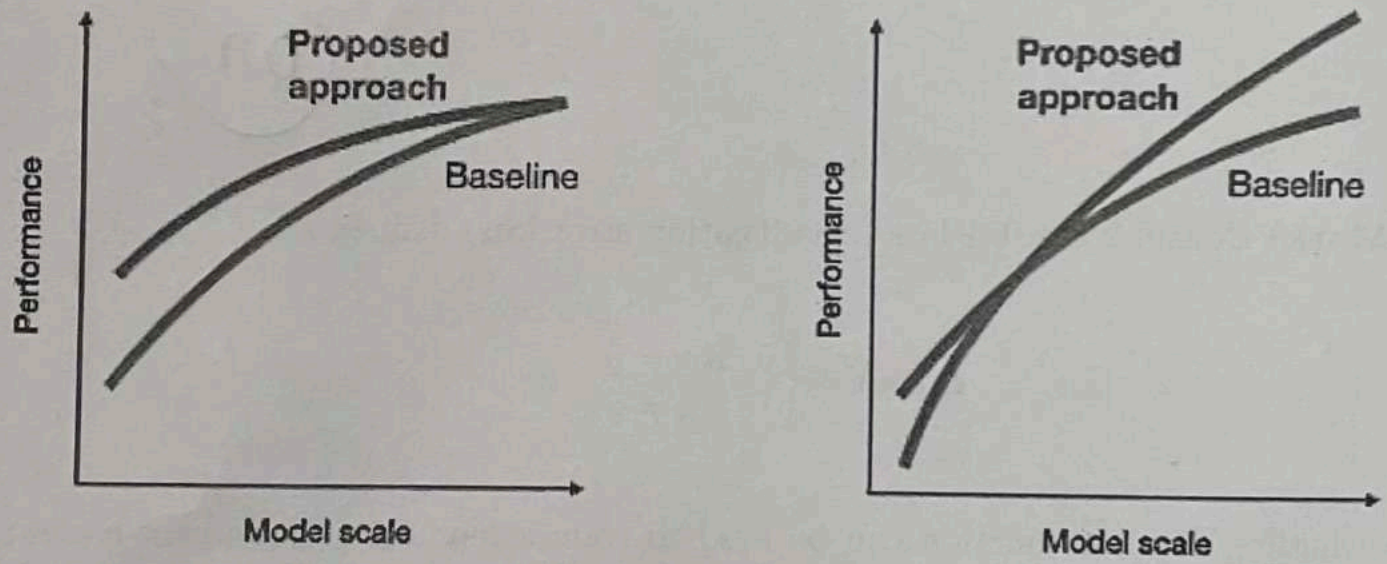
$$\mathcal{L}(y, \hat{y}) = \begin{cases} 0 & \text{if } y = \hat{y} \\ 1 & \text{if } y \neq \hat{y} \end{cases}$$

State whether this loss function can be used to train a model using gradient-based optimization and justify your answer.

Q3. (1 Mark) What is RLHF, and how is it used as a post-training method for aligning large language models?

Q4. (1 Mark) For a Transformer-based LLM, is the forward pass of self-attention permutation-invariant with respect to the input tokens (Yes/No)? Which component makes the forward pass permutation-variant?

Figure: The figure consists of two panels, (a) on the left and (b) on the right. The x-axis represents model scale, while the y-axis denotes model performance for a baseline method and two proposed approaches (a and b).



Q5. (1 Mark) As per the above figure, as AI systems continue to improve, which class of approaches is likely to be more favorable in the future, and why?

Section B

Q1. **(2 Marks)** Explain Model inversion in the context of safety of AI systems and suggest one simple way to do model inversion.

Q2. (2 Marks) Explain how Differential Privacy (DP) mitigates privacy risks in AI systems.

Suppose M is a randomized algorithm. For any two neighboring datasets D and D' that differ in exactly one individual's data, write the mathematical inequality that relates $\Pr(M(D))$ and $\Pr(M(D'))$ for any possible output. Clearly explain how ϵ controls the level of information leakage.

Q3. **(2 Marks)** What is meant by a seam in image processing? Explain the concept of seam carving and describe the outcome after seam carving is applied to an image.

Q4. (2 Marks) Describe briefly how TIBET evaluates bias in Text-to-Image models.

Q5. (2 Marks) What is Latent Adversarial Training (LAT)? Briefly explain how it improves robustness against model tampering attacks.

Section C

Q1. (5 Marks) What is Bias from the Computer Science/Machine Learning perspective? Name any 4 categories of Bias and provide short examples for the same. What are the five major stages in an end-to-end LLM pipeline where bias can be introduced, from data acquisition to real-world use?

Q2. Given a pretrained weight matrix

$$W_0 \in R^{d_{out} \times d_{in}},$$

Low-Rank Adaptation (LoRA) performs task adaptation by learning an additive low-rank update such that

$$W = W_0 + BA,$$

where $B \in R^{d_{out} \times r}$ and $A \in R^{r \times d_{in}}$.

1. Define the uniform quantization step size s for a real-valued tensor $x \in [x_{\min}, x_{\max}]$ represented using b bits. Assuming z denotes the zero-point, write the mathematical expressions for:

   - Forward quantization
   - Dequantization.

2. State the condition imposed on the rank r of the LoRA adapters with respect to the dimensions of the weight matrix. (0.5 marks)

3. What is QLoRA? Provide its mathematical formulation assuming a quantized base model

$$\tilde{W}_0 = Q(W_0).$$

What is the numerical precision used for the adapter matrices in QLoRA? (1 mark)

4. Write the effective weight matrix learned during QLoRA training. Briefly explain why the low-rank adapters BA are able to compensate for the quantization error E. (2 marks)

Q3. Large Language Models (LLMs) are trained using multiple stages, each relying on different notions of ground truth and data annotation.

a) Explain the quality and quantity of data required during the Pretraining, Instruction Tuning and Preference Alignment Stage of LLM training. (1.5 marks)

b) In machine unlearning, training data is partitioned into a *Forget Set* and a *Retain Set*. Define these two sets formally. Provide a concrete example use case where unlearning is required, and construct one well-annotated example that would belong to the Forget Set. (1.5 marks)

c) Unlearning in ML often uses gradient ascent. Let $\mathcal{L}_{retain}$ denote the loss on the Retain Set and $\mathcal{L}_{forget}$ denote the loss on the Forget Set. Design a combined objective function for performing unlearning. (1 mark)

d) What kind of bias can occur if data generated from an LLM is used to further train the LLM? Suggest one potential consequence of it on model behavior. (1 mark)